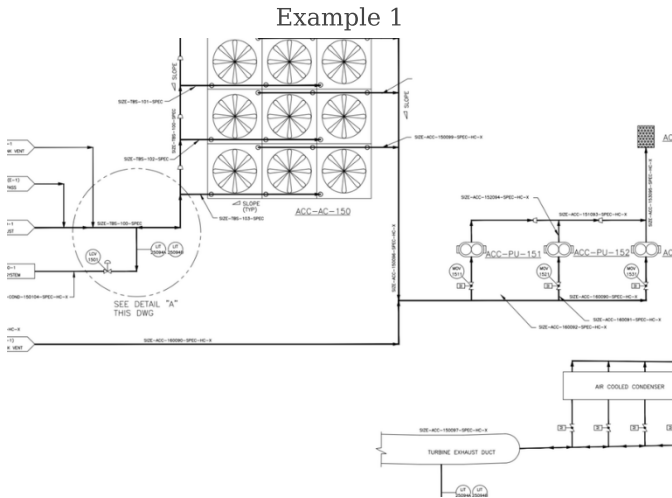
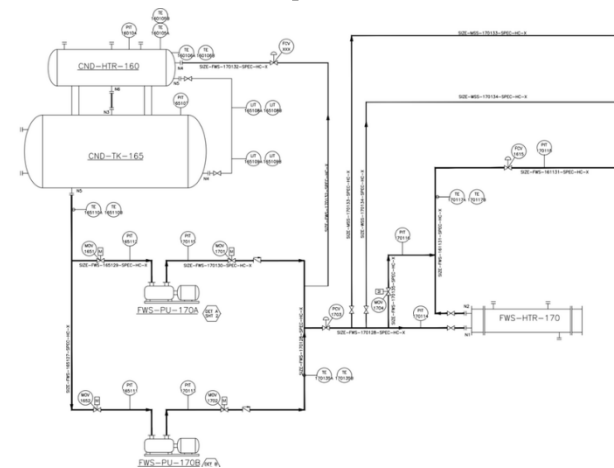


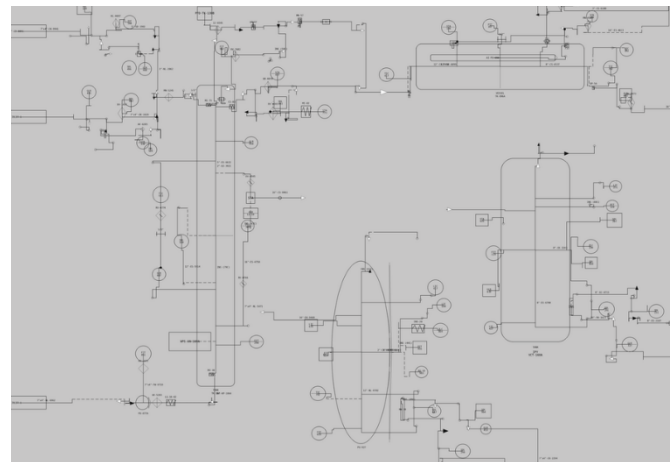
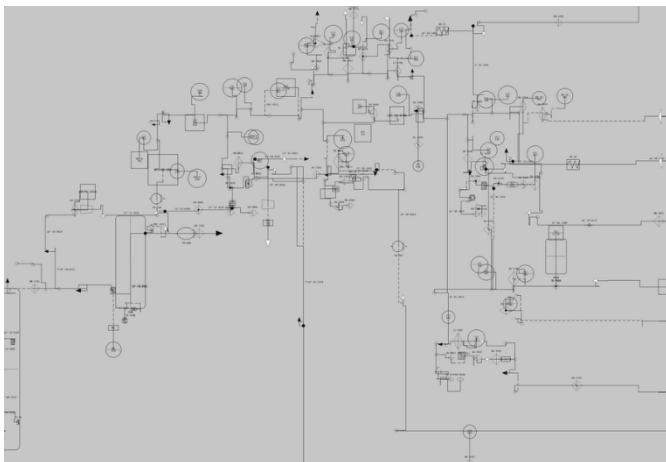
Real P&ID
(OPEN100)



Example 2



Synthetic P&ID
(SynthPID)



Real OPEN100 P&ID Synthetic SynthPID (ours)

Real and synthetic P&IDs share pipe topology and symbol vocabulary but differ in visual style (white vs. ~grey background, rendering). Our model is trained exclusively on SynthPID and evaluated on real OPEN100 P&IDs.